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Plug connector device

Abstract

A plug connector device, in particular an RJ plug connector device, has a plug connector unit which is configured to create a plug connection extending along a plugging direction and which comprises a contact unit with at least two electric plug contacts, wherein the plug connector device includes a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and comprising a connection unit for creating a connection to at least one further element of the cable that is different from a conductor core, and that the plug connector device includes a wiring assistance unit, which is configured for a tool-less establishing of a connection between the wiring block and the contact unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from German Application No. 10 2021 107 183.6, filed on Mar. 23, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) The invention concerns a plug connector device according to the preamble of claim **1**, a plug connector kit for producing a plug connector device according to claim **16** and a method for on-site assembly of an electric cable with a plug connector device according to claim **17**.

(3) From the state of the art, plug connectors are already known which include a plug connector unit comprising a contact unit with a plurality of plug contacts. In some known plug connectors a wiring block for an accommodation of conductor cores of a cable is inserted directly into a housing of the plug connector unit and is connected to the contact unit, a connection of a cable jacket to the housing unit for strain relief and/or a shield contacting of the cable shield being brought about only after a wiring by connecting the wiring block to the contact unit. Herein assembly is disadvantageously encumbered as individual conductor cores may slip from the wiring block while the strain relief and/or the shield contacting are/is established. Moreover, if the strain relief and/or the shield contacting are/is established subsequently, there may be changes in the transmission characteristics, which is a disadvantage. Other known plug connectors comprise a separate unit that includes the wiring block and is configured for a cable preparation realized by establishing a strain relief and/or a shield contacting before a wiring of the contact unit is brought about. However, a disadvantage of such a type of plug connectors is that special tools are required for the wiring, which also makes assembly difficult.

(4) The objective of the invention is in particular to provide a generic device having improved characteristics in regard to assembly. The objective is achieved according to the invention by the features of claim **1** while advantageous implementations and further developments may be gathered from the subclaims.

ADVANTAGES OF THE INVENTION

(5) The invention is based on a plug connector device, in particular an RJ plug connector device, with a plug connector unit which is configured to create a plug connection extending along a plugging direction and which comprises a contact unit with at least two electric plug contacts.

(6) It is proposed that the plug connector device includes a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and comprising a connection unit for creating a connection to at least one further element of the cable, which is different from a conductor core, and that the plug connector device comprises a wiring assistance unit, which is configured for a tool-less establishing of a connection between the wiring block and the contact unit.

(7) Such an implementation allows providing a plug connector device with particularly advantageous characteristics in regard to assembly. Advantageously, in assembly, due to the plug connector unit and the cable receiving unit being realized as separate units, firstly the complete

cable preparation can be made at the cable receiving unit, which means that the conductor cores of the cable can be introduced into the wiring block, after that a shield contacting with the cable shield of the cable and a strain relief of the cable may be realized by creating the connection to the further element of the cable, e. g. the cable jacket, by means of the connection unit of the cable receiving unit. In this way, when this is followed by establishing the connection between the wiring block and the contact unit, it is advantageously possible to effectively prevent the difficulties often occurring with already known plug connector devices, for example individual conductor cores slipping from the wiring block, thus facilitating assembly. It is furthermore ensured that the transmission characteristics do not change when the connection between the wiring block and the contact unit is established, which allows providing a plug connector device having particularly advantageous transmission characteristics, for example a very low degree of cross-talk. The wiring-assistance unit moreover advantageously enables especially easy and tool-less establishing of the connection between the wiring block and the contact unit, thus simplifying assembly even further.

(8) The plug connector device forms at least a portion, in particular a sub-assembly group, of a plug connector, in particular an RJ plug connector, preferably an RJ45 plug connector. The plug connector device may also comprise the entire plug connector. The plug connector may be realized as a plug socket. Preferably the plug connector is realized as a plug. Alternatively, the plug connector may also be realized as a plug connector different than an RJ plug connector, for example as a single pair ethernet plug connector or as a different plug connector that is deemed expedient by someone skilled in the art. The plug connector device is configured for assembly, in particular on-site assembly, of an electric cable, in particular a patch cable. The plug connector device may include a plug connector kit, which comprises at least the plug connector unit and at least the cable receiving unit.

(9) The plug connector unit comprises at least the contact unit and may also comprise further units and/or elements, for example a housing in which the contact unit is arranged at least partially and/or a locking element for a locking and unlocking of the plug connection. The plug connector unit, in particular the housing, the locking element and/or further units and/or elements, with the exception of electrically conductive components, like for example the electric plug contacts of the contact unit, may be made at least partially of a synthetic material. Preferably the plug connector unit, in particular the housing and/or the locking element and/or further units and/or elements, with the exception of electrically conductive components, is made at least partially of a metallic pressure die-casting material, for example a zinc die cast.

(10) The contact unit comprises at least two electric plug contacts. The contact unit may comprise precisely two electric plug contacts. Preferably the contact unit comprises at least four electric plug contacts. The contact unit may comprise at least six, in particular at least eight, electric plug contacts.

(11) The electric cable comprises at least two conductor cores. The electric cable could, for example, be implemented as a single pair ethernet cable having precisely two conductor cores. Preferably the electric cable is implemented as a twisted-pair cable and has at least four, in particular at least six, preferably at least eight, conductor cores, wherein respectively two of the conductor cores are twisted to form a core pair.

(12) Preferably the plugging direction runs parallel to a main extension direction of the plug connector unit. The “main extension direction” of an object is herein a direction to be understood that runs parallel to the longest edge of a smallest geometrical rectangular cuboid completely enclosing the object.

(13) The cable receiving unit comprises at least the wiring block and the connection unit and may also comprise further units and/or elements. The wiring block is configured for an accommodation of the conductor cores of the cable and for a wiring of the contact unit with the conductor cores of the cable. In an assembled state, in particular in a state in which the connection between the wiring block and the contact unit has been brought about by means of the wiring assistance unit, the

wiring block connects the conductor cores of the electric cable in an electrically conductive manner to the contact unit, connecting in particular respectively one conductor core of the electric cable to respectively one electric plug contact of the contact unit. The conductor cores of the cable may in the assembled state be connected non-releasably, for example soldered or crimped, with the contact unit by means of the wiring block. Preferably, the conductor cores of the cable are in the assembled state connected releasably with the contact unit by means of the wiring block. For example, conductor cores of the electric cable may in the assembled state be connected releasably with the contact unit by means of the wiring block via at least one insulation displacement contact (IDC), wherein the conductor cores are in each case pressed by the wiring block with their insulation individually into a so-called insulation-displacement connector of the contact unit in such a way that the insulation is severed and in each case an electrically conductive connection of the conductor cores to the electric plug contact of the contact unit is established. Alternatively the conductor cores of the electric cable could be respectively connected releasably with the contact unit by the wiring block via a so-called insulation piercing contact (IPC), wherein the insulation piercing contact has at least one spike which in the assembled state, in particular by the establishment of the connection between the wiring block and the contact unit, is pressed through the insulation of the conductor core in such a way that in each case an electrically conductive connection of the conductor cores to at least one electric plug contact of the contact unit is brought about.

(14) The connection unit is configured to create the connection to at least one further element of the cable, which is different from a conductor core, and has for this purpose at least one connection element. The connection unit could be configured to create a permanent, in particular non-releasable, connection, for example a substance-to-substance bond, in particular an adhesive connection, to the further element of the cable. Preferably the connection unit is configured to create a releasable, in particular tool-less releasable, connection, in particular a form-fit and/or force-fit connection, to the further element of the cable. The connection element of the connection unit may be realized as a latch element. Preferably the connection element is realized as a clamping element and is configured to engage around the further element of the cable in a form-fit and/or force-fit manner. Preferably the connection unit has a further connection element, which is configured to cooperate with the connection element for creating the connection. The further connection element could, for example, comprise a nut which is configured, for the purpose of creating the connection, to be screwed with a corresponding thread on the connection element.

(15) The further element may, for example, be a cable shield of the electric cable. Preferably the further element is a cable jacket of the electric cable.

(16) The wiring assistance unit is configured at least for a tool-less establishing of the connection between the wiring block and the contact unit. Preferably the wiring assistance unit is moreover configured for a tool-less establishing of a connection between the higher-level units, in particular between the cable receiving unit and the plug connector unit. The wiring assistance unit may be configured to connect the cable receiving unit to the plug connector unit indirectly, namely via the connection between the wiring block and the contact unit. Preferably the wiring assistance unit is configured to connect the cable receiving unit and the plug connector unit to each other directly, namely via at least one further connection, which is different from the connection between the wiring block and the contact unit. Preferably the wiring assistance unit is also configured for a tool-less releasing of the connection between the wiring block and the contact unit.

(17) In the present document numerals like, for example, “first” and “second” prefixed to certain terms only serve for distinguishing objects and/or for allocating objects to one another, without implying any existing total number and/or ranking of the objects. In particular, a “second object” does not necessarily imply the existence of a “first object”.

(18) “Configured” is to mean specifically designed and/or equipped. By an object being configured for a certain function is to be understood that the object fulfills and/or executes said certain

function in at least one application state and/or operation state.

(19) The plug connector unit may comprise a plug socket element for receiving a plug. However, in an advantageous implementation it is proposed that the plug connector unit comprises a plug element for an insertion into a plug socket. Such an implementation allows providing a plug connector device which is part of a plug connector that is embodied as a plug and which has the advantageous characteristics described above.

(20) Beyond this it is proposed that the connection unit comprises a strain relief for a cable jacket of the cable. As a result, assembly is advantageously further improvable. It is in particular possible to prevent inadvertent releasing of the connection between the conductor cores of the cable and the electric plug contacts of the contact unit occurring due to tensile strain during an establishment of the connection between the wiring block and the contact unit. Preferably the strain relief is provided via at least one connection element of the connection unit, which is at the same time configured for creating the connection to the at least one further element of the electric cable that differs from the conductor core, in particular the cable jacket of the electric cable.

(21) It is further proposed that the connection unit comprises a shield contact for contacting a cable shield of the cable. Such an implementation advantageously allows improving assembly even further. It is in particular possible to generate a shield contacting before the connection between the wiring block and the contact unit is brought about, thus advantageously ensuring that transmission characteristics will not change when the contact unit is wired. It would also be conceivable that besides the shield contacting to the cable shield, the shield contact is also configured for a strain relief.

(22) The wiring assistance unit may be connected exclusively to the plug connector unit or could be implemented integrally with the plug connector unit. It would also be conceivable that the wiring assistance unit is connected exclusively to the cable receiving unit or is implemented integrally with the cable receiving unit. However, in an advantageous implementation it is proposed that the wiring assistance unit comprises a first sub-unit, which is part of the plug connector unit, and a second sub-unit, which cooperates with the first sub-unit and is part of the cable receiving unit. This advantageously allows further improving assembly. Moreover, advantageously a particularly simple replacement of the cable guiding unit by a further cable guiding unit also having a second sub-unit will be enabled.

(23) Furthermore it is proposed that the first sub-unit comprises a flap, which is pivotable around a pivot axis extending perpendicularly to the plugging direction and is configured to provide a force acting in the plugging direction for establishing the connection between the wiring block and the contact unit. This advantageously allows improving assembly even further. In particular, tool-less establishing of the connection between the wiring block and the contact unit by simple technical means is enabled. The flap is configured to execute a rotational movement around the pivot axis, the rotational movement being at least partly converted into the force acting in the plugging direction by a cooperation with the second sub-unit.

(24) It is also proposed that the flap has at least one first inner contour, which is configured to cooperate with a first outer contour of the second sub-unit. Such an implementation advantageously allows further improving assembly. The flap may have exactly one first inner contour. However, in an advantageous implementation it is proposed that the flap has at least one second inner contour, which is configured to cooperate with a second outer contour of the second sub-unit. This advantageously allows reducing planar pressure, thus enabling a particularly even force transmission for establishing the connection between the wiring block and the contact unit. Preferably the flap comprises two flap arms, which are oriented perpendicularly to the plugging direction when the connection between the wiring block and the contact unit has been established. Preferably the flap arms are realized mirror-symmetrically to each other and are arranged on two opposite-situated sides of the flap. Preferably the first inner contour and/or the second inner contour are/is respectively arranged on a first one of the flap arms. Preferably the flap has a further

first inner contour, which is configured to cooperate with a further first outer contour of the second sub-unit, and has a further second inner contour, which is configured to cooperate with a further second outer contour of the second sub-unit, the first and second inner contours being respectively arranged on a second one of the flap arms. Preferably, in a view of the flap perpendicularly to the plugging direction, the first inner contour has the shape of a curve which extends, starting with a first region having a course that is substantially parallel to the plugging direction, into a second region having a course that is substantially perpendicular to the plugging direction.

(25) It is further proposed that the first inner contour and the second inner contour are arranged offset from each other in the plugging direction. Such an implementation advantageously allows further improving assembly. In particular, especially even force introduction and force transmission are achievable. Advantageously, it is moreover possible to achieve saving of space, thus providing a particularly compact plug connector device, which may in particular be advantageous in cramped regions, for example in server rooms comprising a plurality of plug connectors that are to be placed in a close-packed manner, as this will also allow increasing plugging density. In addition, it is proposed that the first inner contour is configured to lock the connection between the wiring block and the contact unit. This advantageously allows improving an operator's comfort as inadvertent releasing of the wiring against the plugging direction can be prevented. It is furthermore advantageously possible to provide a particularly compact plug connector device as additional elements for a locking may be done without.

(26) Beyond this it is proposed that the second sub-unit comprises at least one first guiding cam and at least one second guiding cam, which is arranged offset from the first guiding cam in a circumferential direction, the guiding cams being configured to engage in corresponding guiding grooves of the plug connector unit. This advantageously allows further improving assembly. It is in particular possible to prevent jamming during assembly. Moreover, particularly targeted force transmission is advantageously achievable when establishing the connection between the wiring block and the contact unit, such that it is further advantageously possible to obtain a particularly even wiring of all insulation displacement contacts.

(27) It is furthermore proposed that a width extension of the first guiding cam, which extends perpendicularly to the plugging direction, differs from a width extension of the second guiding cam, which extends perpendicularly to the plugging direction. Such an implementation advantageously allows further improving assembly. It is in particular possible to prevent an erroneous assembly, according to the poka-yoke principle, with particularly simple technical means.

(28) In an alternative advantageous implementation it is proposed that the first sub-unit comprises at least one first guiding cam and at least one second guiding cam which is arranged offset from the first guiding cam in a circumferential direction, the guiding cams being configured to engage into corresponding guiding grooves of the second sub-unit. Such an implementation advantageously enables improved assembly. It is in particular possible to prevent jamming during assembly. Moreover, particularly targeted force transmission is advantageously achievable when establishing the connection between the wiring block and the contact unit, such that it is further advantageously possible to obtain a particularly even wiring of all insulation displacement contacts.

(29) Beyond this it is proposed that the guiding cams are arranged off-center with respect to a height extent of the first sub-unit. This advantageously allows further improving assembly. It is in particular possible to achieve an anti-twist protection by simple technical means, thus preventing an erroneous assembly by the poka-yoke principle.

(30) In a further aspect of the invention, which may in particular be considered by its own as well as in a combination with the further aspects of the invention, it is proposed that the contact unit comprises at least four electric plug contacts, a circuit board that is in an assembled state aligned perpendicularly to the plugging direction, and at least four connection lines between the circuit board and the plug contacts, wherein in a view along the plugging direction, connection paths

between respectively two of the connection lines are at least substantially perpendicular to each other. Such an implementation advantageously allows improving transmission technique. It is in particular possible to reduce, preferably to minimize, cross-talk between conductor pairs implemented by respectively two of the connection lines. The connection paths are herein to be understood as imaginary lines running parallel to a main extension plane of the circuit board and in each case through the center points of two of the connection lines which together form a conductor pair. By a “main extension plane” is herein a plane to be understood which is parallel to a largest side surface of a smallest imaginary rectangular cuboid that just still completely encloses the structural unit, and which in particular extends through the center point of the rectangular cuboid. In this document, “at least substantially perpendicular(ly)” is to mean a deviation from a right angle by less than 20° , in particular by less than 15° , advantageously by less than 10° , especially advantageously by less than 7.5° , preferably by less than 5° and particularly preferentially by less than 2.5° . The contact unit may comprise at least two, in particular four, further electric contacts, which are respectively connected to the circuit board via further connection lines. It is conceivable that in a view along the plugging direction, further connection paths between respectively two of the connection lines are at least substantially perpendicular to each other. Preferably the further connection lines are arranged on the circuit board, in at least one direction that is perpendicular to the plugging direction, spaced apart from the at least four connection lines. Preferentially two of the further connection lines that form a further first conductor pair are arranged, in a first direction that is perpendicular to the plugging direction, at a distance to the four connection lines, and two of the further connection lines that form a further second conductor pair are arranged on the circuit board in a second direction, which is opposed to the first direction.

(31) In a further aspect of the invention, which may in particular be considered by its own as well as in a combination with the further aspects of the invention, it is proposed that the contact unit comprises at least four electric plug contacts, a circuit board which is in an assembled state oriented perpendicularly to the plugging direction, and at least one further circuit board, which is in the assembled state oriented parallel to the plugging direction. By such an implementation a transmission technique is advantageously improvable. In particular, cross talk between conductor pairs is reducible, preferably minimizable, if the contact unit comprises a circuit board which is in an assembled state oriented perpendicularly to the plugging direction and at least one further circuit board which is in the assembled state oriented parallel to the plugging direction.

(32) Furthermore, it is proposed that the contact unit comprises at least one EON plug contact, which is in the assembled state pressed into the circuit board. As a result, assembly is advantageously further improvable. In particular, when establishing a connection between the plug contact and the circuit board, soldering may advantageously be dispensed with if the plug contact is embodied as an EON plug contact. In a comparison to soldered connections, this further advantageously allows avoiding a thermal load when establishing the connection between the plug contact and the circuit board. By an “EON plug contact” is a certain type of electric plug contact to be understood which is in the expert language also referred to as a press-fit pin or, in English, as an “Eye-of-the-Needle press-fit connector (EON)”. The EON plug contact comprises a press-fit zone having an elongate hollow space whose shape is in particular similar to a pinhead shape and which is mechanically deformed while being pressed into the circuit board. Thus a high reliability of the connection is advantageously also achievable as during pressing-in of the press-fit zone into a metallized pass-through opening in the circuit board, a deformation of the EON plug contact is realized in the region of the press-fit zone, such that a reliable establishment of the connection can be ensured even in case of fluctuating diameters of the metallized pass-through opening. This further advantageously result in an increased manufacturing tolerance. Further advantageously, due to a high degree of elasticity of the EON plug contact there will be a very small mechanical load of the circuit board during pressing-in. The contact unit may comprise several EON plug contacts, which are pressed into the circuit board in a mutually offset manner. For example, the contact unit

may comprise at least two, in particular at least four, preferably at least six and particularly preferentially at least eight EON plug contacts, which are pressed into the circuit board in a mutually offset manner. Preferably the insulation displacement contacts of the contact unit are also realized as EON plug contacts.

(33) In a further aspect of the invention, which may in particular be considered by its own as well as in a combination with the further aspects of the invention, it is proposed that the plug connector device includes a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and comprising a connection unit for creating a connection to at least one further element of the cable that is different from a conductor core, wherein the cable receiving unit comprises a guiding unit for a guidance of the conductor cores from the connection unit into the wiring block, the guiding unit connecting the connection unit with the wiring block in an angled fashion. Such an implementation advantageously allows improving assembly. In particular, an angled orientation of the electric cable relative to the plugging direction is enabled, thus facilitating—in particular in cramped spaces with a plurality of plug connectors arranged side by side—an establishing of the plug connections. Preferably the guiding unit connects the connection unit with the wiring block in an angled fashion, at an at least substantially right angle, while in particular angles greater or smaller than 90° are also conceivable. Preferably the guiding unit connects the connection unit with the wiring block in an angled fashion in such a way that in an assembled state of the plug connector device, the wiring block is oriented in the plugging direction and the connection unit is oriented at an angle to the plugging direction.

(34) It is further proposed that a contact region of the guiding unit for contacting the wiring block is rotationally symmetrical with respect to the plugging direction. Such an implementation advantageously allows further improving assembly. Preferably the contact region is supported rotatably relative to a further region of the guiding unit, which is connected to the connection unit, and it can be fixed in its orientation relative to the connection unit in at least two, in particular at least three, preferably at least four stages. For this purpose the guiding unit preferably comprises a swivel joint, which connects the contact region to the further region. In this way assembly is even further improvable as an orientation of the electric cable is enabled in different directions in an angled fashion, in particular at least substantially perpendicularly to the plugging direction. The guiding unit preferably comprises a slotted link for force transmission. The slotted link for force transmission is preferably arranged at the further region. Preferably the slotted link for force transmission comprises at least two outer contour elements, which are oriented counter to the plugging direction. Preferentially a number of outer contour elements of the slotted link for force transmission corresponds to a number of stages in which the contact region can be fixed relative to the further region. Preferably each outer contour element has at least one outer contour, whose shape corresponds at least substantially to a shape of the second outer contour of the second sub-unit of the wiring assistance unit.

(35) Beyond this it is proposed that the connection unit comprises a connection flap, which is pivotable around a pivot axis extending perpendicularly to the plugging direction for the purpose of assisting accessibility to a portion of the connection unit. This advantageously allows further improving assembly. It is moreover proposed that the portion of the connection unit comprises a shield contact, which is pivotable around a further pivot axis extending perpendicularly to the plugging direction for the purpose of assisting with a shield contacting. This advantageously allows further improving assembly. In particular, shield contacting is facilitated. The shield contact is preferably pivotable into at least two positions, wherein the shield contact is in a first position oriented in the direction of the plugging direction and is in a second position oriented at an angle to the plugging direction. Preferably, in the second position an angle between the shield contact and the plugging direction is at least substantially equivalent to an angle at which the guiding unit connects the wiring block to the connection unit.

(36) In a further advantageous implementation it is proposed that the guiding unit is realized in a

multi-part fashion and comprises a guiding base body, which is oriented parallel to the plugging direction, as well as at least two deflection elements for an angled connection of the connection unit to the wiring block. Such an implementation advantageously allows further improving assembly. It is moreover advantageously possible to improve flexibility, in particular as a flexible adaption of the connection unit to a desired spatial orientation between the cable and the plugging direction is particularly easily possible.

(37) It is also proposed that the deflection elements are realized as corresponding halfshells and are configured for establishing a form-fit connection to the guiding base body. This advantageously allows further improving assembly. Advantageously a toolless and particularly simple establishment of an angled connection of the connection unit to the wiring block is enabled.

(38) Furthermore it is proposed that the plug connector device comprises a shield contact, which is rotatably connected to the guiding base body and is latchable in at least four positions. Such an implementation advantageously allows further improving flexibility, in particular as a flexible adaption of the shield contacting to a desired spatial orientation between the cable and the plugging direction is particularly easily possible. Preferably the shield contact is rotatable with the guiding base body by an angle of 360° . Preferably the shield contact is releasably connected to the guiding base body. The guiding base body advantageously has at least four latching grooves, which are arranged offset along its circumferential direction, wherein the shield contact is configured to latch into one of the latching grooves depending on a desired orientation. The shield contact is preferably pivotable around a pivot axis that extends perpendicularly to the plugging direction, and is in particular pivotable between an orientation parallel to the plugging direction and an orientation perpendicular to the plugging direction.

(39) In a further aspect of the invention, which may in particular be considered by its own as well as in a combination with the further aspects of the invention, it is proposed that the plug connector device includes a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable, and comprising a connection unit for creating a connection to at least one further element of the cable that is different from a conduction core, wherein the connection unit comprises at least one connection element with a plurality of lamellae, which are configured to engage around the further element of the cable along a circumferential direction and which are oriented counter to the plugging direction and configured for an optical indication. Such an implementation advantageously allows providing a plug connector device having advantageous characteristics regarding assembly. As the lamellae are oriented counter to the plugging direction, an optical indication concerning a state of the connection to the at least one element that differs from a conductor core is achievable with particularly simple technical means. It is advantageously possible to easily prevent damaging of the connection element, for example due to over-tightening when creating the connection. Preferably the connection element is implemented as an elastic element, which is repeatedly deformable without being mechanically damaged or destroyed, and which following a deformation autonomously seeks to re-assume its original shape.

(40) In a further aspect of the invention, which may in particular be considered by its own as well as in a combination with the other aspects of the invention, it is proposed that the plug connector device includes a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and a connection unit for creating a connection to at least one further element of the cable which is different from a conductor core, wherein the connection unit comprises at least one sleeve-like connection element, which is configured to engage around the further element of the cable along a circumferential direction and to generate a resistance against a tensile load of the cable counter to the plugging direction. Such an implementation advantageously allows improving user-friendliness. It is in particular possible to counteract an undesired releasing of the conductor cores out of the wiring block in case of a tensile load of the cable counter to the plugging direction. Preferably the sleeve-like connection element is embodied as a squeeze element. Preferably the connection unit comprises a connection element and a further connection

element, which act together for creating the connection to at least one further element of the cable that is different from a conductor core. For example, the connection element may comprise a thread and the further connection element may comprise a nut that is configured to be screwed with the thread of the connection element for the purpose of creating the connection. When the connection between the further element of the cable and the further connection element is created, the sleeve-like connection element is arranged between the further element of the cable and the further connection element and, when the further connection element is connected to the connection element, the sleeve-like connection element is pressed into the further element of the cable, for example the cable jacket of the cable, such that a considerable resistance against a pulling-out of the cable out of the cable receiving unit is generated. Preferably the sleeve-like connection element is realized as an elastic element, which is repeatedly deformable without being mechanically damaged or destroyed, and which following a deformation autonomously seeks to reassume its original shape. Preferably the sleeve-like connection element comprises several recesses which are arranged offset to each other along its circumferential direction, thus advantageously allowing even further increasing a resistance against a pulling-out of the cable out of the cable receiving unit.

(41) The invention further concerns a plug connector kit for producing a plug connector device according to one of the above-described implementations with the plug connector unit and with the cable receiving unit. Such a plug connector kit is in particular distinguished by its advantageous characteristics regarding an assembly for producing the plug connector device. The plug connector kit may advantageously comprise the plug connector unit and at least two cable receiving units, which may be realized at least partially differently from each other, in particular according to respective aspects and/or combinations of the above-described implementations of the plug connector device. As a result, a particularly high degree of flexibility is achievable. It is also conceivable that the plug connector kit comprises at least two different plug connector units differing from each other, for example, regarding a number of electric plug contacts of the contact unit. As a result, a particularly high degree of flexibility is achievable. The plug connector kit may also comprise at least one electric cable.

(42) Furthermore a method is proposed for a field assembly of an electric cable with a plug connector device according to one of the previously described implementations, wherein at least one conductor core of the electric cable is connected to at least one electric plug contact of the contact unit. Such a method advantageously allows obtaining particularly simple, quick and secure field assembly by which furthermore, in particular in commercial applications with a plurality of cables that are to be assembled, time-saving and thus cost-saving are advantageously achievable.

(43) The plug connector device according to the invention as well as the plug connector kit shall here not be limited to the applications and implementations described above. In particular, for the purpose of fulfilling a functionality that is described here, the plug connector device according to the invention and/or the plug connector kit may comprise a number of individual elements, components and units that differs from a number that is given here.

Description

DRAWINGS

(1) Further advantages will become apparent from the following description of the drawings. In the drawings four exemplary embodiments of the invention are illustrated. The drawings, the description and the claims contain a plurality of features in combination. Someone skilled in the art will purposefully also consider the features separately and will find further expedient combinations.

(2) It is shown in: a plug connector device with a plug connector unit and a cable receiving unit, in a schematic perspective view, an electric cable in a schematic sectional view, the plug connector device in a schematic exploded view, a plug connector kit for producing the plug connector device

with the plug connector unit and the cable receiving unit, in a schematic perspective view, a further schematic perspective view of the plug connector kit of FIG. 4, a contact unit of the plug connector unit with a circuit board and with a plurality of electric plug contacts, in two schematic views, a schematic flow chart of a method for a field assembly of the electric cable with the plug connector unit, a further exemplary embodiment of a plug connector device with a plug connector unit and a cable receiving unit, in a schematic perspective view, the cable receiving unit of the plug connector device of FIG. 8, in a schematic perspective view, a further exemplary embodiment of a plug connector device with a plug connector unit and a cable receiving unit, in a schematic exploded view, the plug connector unit of FIG. 10 in a schematic perspective view, a contact unit of the plug connector unit of the exemplary embodiment of FIGS. 10 and 11 with a circuit board, a plurality of electric plug contacts and a further circuit board, in three schematic views, a further exemplary embodiment of a plug connector device with a plug connector unit and a cable receiving unit, in a schematic view, and the cable receiving unit of the plug connector device of the exemplary embodiment of FIG. 13, in a schematic exploded view.

DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

- (3) FIG. 1 illustrates a plug connector device **10a** in a schematic perspective view. The plug connector device **10a** is configured to create a plug connection that extends along a plugging direction **14a**.
- (4) The plug connector device **10a** comprises a plug connector unit **12a**. The plug connector unit **12a** comprises a contact unit **16a**. The contact unit **16a** comprises at least two electric plug contacts **18a**, **102a** (see FIG. 6).
- (5) The plug connector unit **12a** comprises a plug element **34a** for an insertion into a plug socket (not shown).
- (6) The plug connector device **10a** comprises a cable receiving unit **20a** for receiving an electric cable **26a** (see FIG. 2).
- (7) If there are a plurality of objects, only one of these objects is given a reference numeral in the figures.
- (8) FIG. 2 shows a schematic sectional view of the electric cable **26a**. The electric cable **26a** is embodied as a twisted-pair cable and is configured for a data transmission. The electric cable **26a** comprises a plurality of conductor cores **24a**. In the present case the electric cable **26a** comprises eight conductor cores **24a** in total. Each of the conductor cores **24a** is encompassed by a core sheath **130a**. Respectively two of the conductor cores **24a** are twisted to form a core pair of the electric cable **26a**. The electric cable **26a** comprises a cable shield **42a**. The cable shield **42a** encompasses the conductor cores **24a** along a circumferential direction **98a**. The electric cable **26a** comprises a cable jacket **38a**. The cable jacket **38a** encompasses the cable shield **42a** along the circumferential direction **98a**.
- (9) FIG. 3 shows the plug connector device **10a** in a schematic exploded view. The plug connector unit **12a** comprises a housing **36a**. The housing **36a** is made of a metallic die-cast material and is configured to accommodate the contact unit **16a** and the plug element **34a**.
- (10) The cable receiving unit **20a** comprises a wiring block **22a** for receiving conductor cores **24a** of the electric cable **26a** (see FIG. 2). The cable receiving unit **20a** comprises a connection unit **28s** for establishing a connection to at least one further element **30a** of the cable **26a**. The wiring block **22a** is releasably connected to the connection unit **28a**, namely via a plug connection. In the present case the further element **30a** is the cable jacket **38a** of the electric cable **26a** (see FIG. 2).
- (11) The connection unit **28a** comprises a connection base body **132a**. The connection base body **132a** has a thread **134a**. The connection unit **28a** comprises at least one connection element **94a**. The connection element **94a** comprises a plurality of lamellae **96a**. The lamellae **96a** are configured to engage around the further element **30a**, in the present case the cable jacket **38a**, of the electric cable **26a** along the circumferential direction **98a** (see FIG. 2). When the connection to the further element **30a** of the electric cable **26a** has been created by the connection unit **28a**, the

lamellae **96a** of the connection element **94a** are oriented counter to the plugging direction **14a**. The lamellae **96a** are configured for an optical indication. The connection unit **28a** includes a further connection element **136a**, which comprises a nut **138a** corresponding to the thread **134a**. In order to create the connection to the further element **30a** of the electric cable **26a**, the connection element **94a** is slid onto the electric cable **26a** together with the further connection element **136a** in such a way that the lamellae **96a** are oriented counter to the plugging direction **14a**. Then the nut **138a** is screwed onto the thread **134a**, wherein the lamellae **96a** are pressed toward the cable jacket **38a** and the connection is created. As the lamellae **96a** are oriented counter to the plugging direction **14a**, an optical indication of the strength of the connection is given.

(12) The connection unit **28a** comprises a strain relief for the cable jacket **38a** (see FIG. 2) of the electric cable **26a**. The strain relief is brought about via the connection unit **28a** by means of the connection element **94a** and the further connection element **136a**.

(13) The connection unit **28a** comprises a shield contact **40a** for a contacting of the cable shield **42a** (see FIG. 2) of the electric cable **26a**. For a shield contacting, the shield contact **40a** is connected to the cable shield **42a**. By the connection brought about by the connection element **94a** and the further connection element **136a**, the shield contacting is additionally safeguarded from getting released counter to the plugging direction **14a**.

(14) The plug connector device **10a** comprises a wiring assistance unit **32a**. The wiring assistance unit **32a** is configured for tool-less establishing a connection between the wiring block **22a** and the contact unit **16a**.

(15) In FIGS. 4 and 5 a plug connector kit **100a** for creating the plug connector device **10a** is illustrated in two different perspective schematic views. The plug connector kit **100a** comprises the plug connector unit **12a** and the cable receiving unit **20a**. The plug connector device **10a** can be assembled from the plug connector kit **100a**, wherein an especially simple and intuitive assembly is brought about due to the features of the wiring assistance unit **32a**, which will be described below.

(16) The wiring assistance unit **32a** comprises a first sub-unit **44a**. The first sub-unit **44a** is part of the plug connector unit **12a**. The wiring assistance unit **32a** comprises a second sub-unit **46a** which cooperates with the first sub-unit **44a**. The second sub-unit **46a** is part of the cable receiving unit **20a**.

(17) The first sub-unit **44a** comprises a flap **48a**. The flap **48a** is pivotable around a pivot axis **50a** that extends perpendicularly to the plugging direction **14a**. The flap **48a** is configured to provide a force that acts in the plugging direction **14a** for establishing the connection between the wiring block **22a** and the contact unit **16a**. The flap **48a** has at least one first inner contour **52a**. The first inner contour **52a** is configured to cooperate with a first outer contour **54a** of the second sub-unit **46a**. The flap **48a** has a second inner contour **56a**. The second inner contour **56a** is configured to cooperate with a second outer contour **58a** of the second sub-unit **46a**. The first inner contour **52a** and the second inner contour **58a** are arranged offset from each other in the plugging direction **14a**. Accordingly the first outer contour **54a** and the second outer contour **58a** are also arranged offset from each other in the plugging direction **14a**.

(18) The first inner contour **52a** is configured for locking the connection between the wiring block **22a** and the contact unit **16a**. When the connection has been established, the inner contour **52a** locks the wiring block **22a** with the contact unit **16a** counter to the plugging direction **14a**.

(19) The second sub-unit **46a** comprises at least one first guiding cam **60a**. The first outer contour **54a** is formed by the guiding cam **60a**. The second sub-unit **46a** comprises at least one second guiding cam **64a**. The second guiding cam **64a** is arranged offset from the first guiding cam **60a** in a circumferential direction **98a**. The guiding cams **60a**, **64a** are configured to engage in corresponding guiding grooves **62a**, **66a** of the plug connector unit **12a**. A first guiding groove **62a** of the plug connector unit **12a** corresponds to the first guiding cam **60a** of the second sub-unit **46a**. A second guiding groove **66a** of the plug connector unit **12a** corresponds to the second guiding cam **64a** of the second sub-unit **46a**. In the tool-less establishing of the connection between the

wiring block **22a** and the contact unit **16a**, the guiding cams **60a**, **64a** engage in the respectively corresponding guiding grooves **62a**, **66a** of the plug connector unit **12a** and are slid into the guiding grooves **62a**, **66a** in the plugging direction **14a** by the force provided via the flap **48a**.

(20) A width extension **68a** of the first guiding cam **60a**, which extends perpendicularly to the plugging direction **14a**, differs from a width extension **70a** of the second guiding cam **64a**, which extends perpendicularly to the plugging direction **14a**. In the present case the width extension **70a** of the second guiding cam **64a** is greater than the width extension **68a** of the first guiding cam **60a**. The difference between the width extension **68a** and the width extension **70a** permits preventing an erroneous assembly when the connection between the wiring block **22a** and the contact unit **16a** is established.

(21) FIG. 6 shows the contact unit **16a** of the plug connector device **10a** in two schematic illustrations. In a lefthand illustration the contact unit **16a** is depicted in a schematic perspective manner. The contact unit **16a** comprises at least four electric plug contacts **18a**, **102a**, **104a**, **106a**. The contact unit **16a** comprises a circuit board **72a**. In an assembled state of the contact unit **16a** within the plug connector unit **12a**, the circuit board **72a** is oriented perpendicularly to the plugging direction **14a**. The contact unit **16a** comprises at least four connection lines **74a**, **76a**, **78a**, **80a** between the circuit board **72a** and the electric plug contacts **18a**, **102a**, **104a**, **106a**, wherein in the assembled state each of the connection lines **74a**, **76a**, **78a**, **80a** connects respectively one of the electric plug contacts **18a**, **102a**, **104a**, **106a** to the circuit board **72a** in an electrically conductive manner.

(22) In the present case the contact unit comprises eight electric plug contacts in total, namely the four electric plug contacts **18a**, **102a**, **104a**, **106a** and four further electric plug contacts **108a**, **110a**, **112a**, **114a**. The further electric plug contacts **108a**, **110a**, **112a**, **114a** are in each case electrically conductively connected to the circuit board **72a** via further connection lines **116a**, **118a**, **120a**, **122a** of the contact unit **16a**.

(23) The contact unit **16a** comprises a plurality of insulation displacement contacts **124a**, which are connected to the circuit board **72a**. In the present case the contact unit **16a** comprises eight insulation displacement contacts **124a** in total. In the assembled state the insulation displacement contacts **124a** are oriented counter to the plugging direction **14a**. When the connection between the wiring block **22a** and the contact unit **16a** is established via the wiring assistance unit **32a**, an electrically conductive connection is created between the conductor cores **24a** of the electric cable **26a** and the electric plug contacts **18a**, **102a**, **104a**, **106a**, respectively the further electric plug contacts **108a**, **110a**, **112a**, **114a** via the insulation displacement contacts **124a**.

(24) A righthand-side schematic illustration of FIG. 6 shows the circuit board **72a** in a lateral view in a view direction along the plugging direction **14a**. In a view along the plugging direction **14a**, connection paths **126a**, **128a** between respectively two of the connection lines **74a**, **76a**, **78a**, **80a** are at least substantially perpendicular to each other. In the present case a first connection path **126a**, which extends respectively diagonally through the connection points in which the connection line **74a** and the connection line **76a** are connected to the circuit board **72a**, is at least substantially perpendicular to a second connection path **128a**, which extends respectively diagonally through the connection points in which the connection line **78a** and the connection line **80a** are connected to the circuit board **72a**. In an operating state of the plug connector device **10a**, the electric plug contact **18a** and the electric plug contact **102a**, which are connected to the circuit board **72a** via the connection line **74a** and the connection line **76a**, form a first conductor pair. In the operating state of the plug connector device **10a**, the electric plug contact **104a** and the electric plug contact **106a**, which are connected to the circuit board **72a** via the connection line **78a** and the connection line **80a**, form a second conductor pair. Due to the above-described arrangement of the connection lines **74a**, **76a**, **78a**, **80a** with connection paths **126a**, **128a** that are at least substantially perpendicular to each other, cross-talk between the first conductor pair and the second conductor pair can be reduced in the operating state of the plug connector device **10a**.

(25) FIG. 7 shows a schematic flow chart of a method for a field assembly of the electric cable **26a** (see FIG. 2) with the plug connector device **10a**. In the method at least one conductor core **24a** of the electric cable **26a** is connected to at least one electric plug contact **18a**, **102a**, **104a**, **106a** of the contact unit **16a**. The method comprises at least two method steps. In a first method step **144a** a cable preparation of the electric cable **26a** is carried out; herein firstly the conductor cores **24a** of the electric cable **26a** are introduced into the wiring block **22a** (see FIG. 3). After this the shield contacting is brought about between the shield contact **40a** (see FIG. 3) and the cable shield **42a** (see FIG. 2) in the first method step **144a**. In the first method step **144a** then the connection to the at least one further element **30a**, which is different from a conductor core **24a**, is created by means of the connection unit **28a** (see FIG. 3), thus obtaining the strain relief of the cable **26a**. In a second method step **146a** of the method, the connection between the wiring block **22a** and the contact unit **16a** is established in a tool-less manner; this is done via the wiring assistance unit **32a** (see FIGS. 4 and 5). Herein the conductor cores **24a** are clamped into the insulation displacement contacts **124a** of the contact unit **16a** (see FIG. 6) in such a way that the core sheaths **130a** encompassing the conductor cores **24a** (see FIG. 2) are severed. In this way the conductor cores **24a** are connected in an electrically conductive manner to the corresponding plug contacts **18a**, **102a**, **104a**, **106a** by the insulation displacement contacts **124a** via the circuit board **72a** and via the connection lines **74a**, **76a**, **78a**, **80a**.

(26) In FIGS. 8 to 14 three further exemplary embodiments of the invention are shown. The following descriptions and the drawings are essentially limited to the differences between the exemplary embodiments, wherein regarding components having the same denomination, in particular regarding components having the same reference numerals, the drawings and/or the description of the other exemplary embodiments, in particular of FIGS. 1 to 7, may principally also be referred to. For distinguishing between the exemplary embodiments, the letter a has been added to the reference numerals of the exemplary embodiment of FIGS. 1 to 7. In the exemplary embodiments of FIGS. 8 to 14 the letter a has been substituted by the letters b to d.

(27) FIG. 8 shows a further exemplary embodiment of a plug connector device **10b** in a schematic illustration. The plug connector device **10b** comprises a plug connector unit **12b** for creating a plug connection that extends along a plugging direction **14b**. The plug connector device **10b** further comprises a cable receiving unit **20b**.

(28) The plug connector device **10b** differs from the plug connector device **10a** of the preceding exemplary embodiment essentially in regard to an implementation of the cable receiving unit **20b**. Regarding a construction and a functionality of the plug connector unit **12b**, the above description of the plug connector device **10a** of the preceding exemplary embodiment may be referred to.

(29) The cable receiving unit **20b** comprises a wiring block **22b** for an accommodation of conductor cores of an electric cable (not shown here, see FIG. 2). The wiring block **22b** is realized substantially identically to the wiring block **22a** of the cable receiving unit **20a** of the preceding exemplary embodiment, which is why at this point the above description of FIGS. 1 to 7 is referred to in regard to the construction and the functionality of the wiring block **22b**.

(30) The cable receiving unit **20b** comprises a connection unit **28b** for creating a connection to at least one further element (not shown here, see FIG. 2) of the electric cable.

(31) FIG. 9 shows the cable receiving unit **20b** in a schematic perspective illustration.

(32) The cable receiving unit **20b** comprises a guiding unit **82b** for guiding the conductor cores of the electric cable from the connection unit **28b** into the wiring block **22b**. The guiding unit **82b** connects the connection unit **28b** to the wiring block in an angled fashion. In the present case, the guiding unit **82b** connects the connection unit **28b** to the wiring block **22b** at an angle of at least substantially 90°, wherein angles greater or smaller than 90° would principally also be conceivable.

(33) The guiding unit **82b** comprises a contact region **84b** for contacting the wiring block **22b**. The contact region **84b** is realized so as to be rotationally symmetrical with respect to the plugging direction **14b**. The guiding unit **82b** comprises a slotted link for force transmission **140b**. The

slotted link for force transmission **140b** comprises a total of four outer contour elements **142b**, which are arranged on four sides of the guiding unit **82b**. The outer contour elements **142b** each have an outer contour having a shape that is at least substantially identical to the shape of the first inner contour **52a** of the second sub-unit **46a** of the wiring assistance unit **32a** of the plug connector device **10a** of the preceding exemplary embodiment.

(34) The connection unit **28b** comprises a connection flap **86b**. To support accessibility to a portion **90b** of the connection unit **28b**, the connection flap **86b** is pivotable around a pivot axis **88b** that extends perpendicularly to the plugging direction **14b**.

(35) The portion **90b** comprises a shield contact **40b**. To support a shield contacting of the electric cable, the shield contact **40b** is pivotable around a further pivot axis **92b** that extends perpendicularly to the plugging direction **14b**. For a shield contacting with a cable shield of the electric cable (not shown here, see FIG. 2), the shield contact **40b** is pivotable around the further pivot axis **92b** such that the shield contact **40b** is oriented substantially parallel to the plugging direction **14b**. When the shield contacting has been brought about, the shield contact **40b**, together with the electric cable, can be pivoted around the further pivot axis **92b** once again such that the shield contact **40b** is oriented substantially perpendicularly to the plugging direction **14b**.

(36) FIG. 10 shows a further exemplary embodiment of a plug connector device **10c** in a schematic exploded view. The plug connector device **10c** comprises a plug connector unit **12c** for establishing a plug connection that extends along a plugging direction **14c**. The plug connector unit **12c**

includes a contact unit **16c**. The contact unit **16c** comprises at least two electric plug contacts **18c**, **102c** (see FIG. 12). The plug connector device **10c** further comprises a cable receiving unit **20c**.

(37) The cable receiving unit **20c** comprises a wiring block **22c** for an accommodation of conductor cores of an electric cable (not shown here, see FIG. 2). The cable receiving unit **20c** comprises a connection unit **28c** for creating a connection to a further element (not shown) of the cable that is different from a conductor core, for example a cable jacket of the cable. The wiring block **22c** is releasably connected to the connection unit **28c**, namely via a plug connection.

(38) The connection unit **28c** includes a connection base body **132c**. The connection base body **132c** comprises a thread **134c**. The connection unit **28c** comprises at least one connection element **94c**. In contrast to the connection element **94a** of the connection unit **28a** of the plug connector device **10a** of the first exemplary embodiment, the connection element **94c** is realized as a sleeve-like connection element **94c**. The sleeve-like connection element **94c** is configured to engage around the further element of the cable along a circumferential direction and to generate a resistance against a strain load of the cable counter to the plugging direction **14c**. The connection unit **28c** comprises a further connection element **136c**, which comprises a nut **138c** that corresponds to the thread **134c**. When the nut **138c** is screwed onto the thread **134c**, the sleeve-like connection element **94c** is pressed onto the further element of the cable, e. g. the cable jacket of the cable, thus generating a considerable resistance against the cable getting pulled out counter to the plugging direction **14c**. The sleeve-like connection element **94c** comprises several recesses **182c**, which are arranged offset from one another along the circumferential direction of the sleeve-like connection element **94c**. The resistance against the cable getting pulled out counter to the plugging direction **14c** is further enforced by the recesses **182c**.

(39) FIG. 11 shows the plug connector device **10c** in a schematic perspective view. The plug connector device **10c** comprises a wiring assistance unit **32c**. The wiring assistance unit **32c** is configured for a tool-less establishing of a connection between the wiring block **22c** and the contact unit **16c**.

(40) The wiring assistance unit **32c** comprises a first sub-unit **44c**. The first sub-unit **44c** is part of the plug connector unit **12c**. The wiring assistance unit **32c** comprises a second sub-unit **46c** which cooperates with the first sub-unit **44c**. The second sub-unit **46c** is part of the cable receiving unit **20c**.

(41) The first sub-unit **44c** comprises a flap **48c**. The flap **48c** is pivotable around a pivot axis **50c**

that extends perpendicularly to the plugging direction **14c**. The flap **48c** is configured to provide a force acting in the plugging direction **14c** for establishing the connection between the wiring block **22c** and the contact unit **16c**. The flap **48c** has at least one first inner contour **52c**. The first inner contour **52c** is configured to cooperate with a first outer contour **54c** of the second sub-unit **46c**.

(42) The first inner contour **52c** is configured to lock the connection between the wiring block **22c** and the contact unit **16c**. When the connection has been established, the first inner contour **52c** locks the wiring block **22c** with the contact unit **16c** counter to the plugging direction **14c**.

(43) The first sub-unit **44c** comprises at least one first guiding cam **60c** and at least one second guiding cam **64c**. The second guiding cam **64c** is arranged offset from the first guiding cam **60c** in a circumferential direction. The guiding cams **60c**, **64c** are configured to engage in corresponding guiding grooves **62c**, **66c** of the second sub-unit **46c**. A first guiding groove **62c** of the second sub-unit **46c** corresponds to the first guiding cam **60c** of the first sub-unit **44c**. A second guiding groove **66c** of the second sub-unit **46c** corresponds to the second guiding cam **64c** of the first sub-unit **44c**. In the present exemplary embodiment, the arrangement of the guiding cams **60c**, **64c** and the guiding grooves **62c**, **66c** are switched relative to the arrangement in the first exemplary embodiment. The guiding grooves **62c**, **66c** are in the present case arranged, on opposite-situated sides, in the wiring block **22c** and the connection base body **132c**, which are part of the second sub-unit **46c**. The guiding cams **60c**, **64c** are arranged off-center with respect to a height extent **148c** of the first sub-unit **44c**. As a result, an anti-twist protection is enabled in assembly and a correct connection between the wiring block **22c** and the contact unit **16c** is ensured. In a tool-less establishment of the connection between the wiring block **22c** and the contact unit **16c**, the guiding cams **60c**, **64c** of the first sub-unit **44c** engage in the respectively corresponding guiding grooves **62c**, **66c** of the second sub-unit **46c** and are slid into the guiding grooves **62c**, **66c** via the force provided by the flap **48c** counter to the plugging direction **14c**.

(44) FIG. 12 shows the contact unit **16c** of the plug connector device **10c** in three schematic illustrations. In a left-hand illustration the contact unit **16c** is shown in a schematic perspective view. The contact unit **16c** comprises at least four electric plug contacts **18c**, **102c**, **104c**, **106c**. The contact unit **16c** comprises a circuit board **72c**. In an assembled state of the contact unit **16c** within the plug connector unit **12c**, the circuit board **72c** is oriented perpendicularly to the plugging direction **14c**. The contact unit **16c** comprises at least one further circuit board **150c**, which is in the assembled state oriented parallel to the plugging direction **14c**. The circuit board **72c** and the further circuit board **150c** are in each case embodied as printed circuit boards.

(45) The contact unit **16c** comprises, in addition to the four electric plug contacts **18c**, **102c**, **104c**, **106c**, four further electric plug contacts **108c**, **110c**, **112c**, **114c**. The electric plug contacts **18c**, **102c**, **104c**, **106c** and the further electric plug contacts **108c**, **110c**, **112c**, **114c** are in the assembled state plugged—in an alternating sequence—into the further circuit board **150c** on an upper side. The contact unit **16c** moreover comprises four further plug contacts **152c**, **154c**, **156c**, **158c**, which are in the assembled state plugged into the further circuit board **150c** on its upper side opposite the electric plug contacts **18c**, **102c**, **104c**, **106c**. Respectively one of the electric plug contacts **18c**, **102c**, **104c**, **106c** is electrically conductively connected to respectively one of the further plug contacts **152c**, **154c**, **156c**, **158c** via a conductor path in the further circuit board **150c**. The contact unit **16c** furthermore comprises four further plug contacts **160c**, **162c**, **164c**, **166c**, which are in the assembled state plugged into the further circuit board **150c** on its underside. Respectively one of the further electric plug contacts **108c**, **110c**, **112c**, **114c** is electrically conductively connected to respectively one of the further plug contacts **160c**, **162c**, **164c**, **166c** via a conductor path in the further circuit board **150c**.

(46) The contact unit **16c** comprises at least one EON plug contact **168c**, which is in the assembled state pressed into the circuit board **72c**. In the present case, the further plug contacts **152c**, **154c**, **156c**, **158c** and the further plug contacts **160c**, **162c**, **164c**, **166c** are in each case realized as EON plug contacts **168c**.

(47) A middle schematic illustration of FIG. 12 shows the circuit board 72c in a lateral view in a view direction counter to the plugging direction 14c. The circuit board 72c comprises sixteen metallized pass-through openings 170c. The further plug contacts 152c, 154c, 156c, 158c, 160c, 162c, 164c, 166c, realized as EON plug contacts 168c, in each case have press-fit zones 184c. In a right-hand illustration of FIG. 12 the further plug contact 166c with a press-fit zone 184c is schematically shown by way of example. In the assembled state, the press-fit zones 184c are pressed into the circuit board 72c, thus connecting the further circuit board 150c to the circuit board 72c.

(48) The contact unit 16c comprises a plurality of insulation displacement contacts 124c, which are connected to the circuit board 72c. The insulation displacement contacts 124c are also realized as EON plug contacts 168c and have a press-fit zone 184c for plugging into one of the metallized pass-through openings 170c. In the present case, the contact unit 16c comprises eight insulation displacement contacts 124c in total. In the right-hand illustration of FIG. 12 one of the insulation displacement contacts 124c is shown exemplarily. Regarding the functionality of the insulation displacement contacts 124c, the above description of FIG. 6 concerning the contact unit 16a of the first exemplary embodiment may be referred to.

(49) FIG. 13 shows a further exemplary embodiment of a plug connector device 10d in a schematic view. The plug connector device 10d comprises a plug connector unit 12d for creating a plug connection that extends along a plugging direction 14d. The plug connector device 10d further comprises a cable receiving unit 20d.

(50) The plug connector device 10d differs from the plug connector devices 10a-c of the preceding exemplary embodiments essentially in regard to an implementation of the cable receiving unit 20d.

(51) The cable receiving unit 20d comprises a wiring block 22d (see FIG. 14) for an accommodation of conductor cores of an electric cable (not shown here, see FIG. 2). The wiring block 22d is implemented substantially identically to the wiring block 22c of the cable receiving unit 20c of the preceding exemplary embodiment.

(52) The cable receiving unit 20d comprises a connection unit 28d for creating a connection to at least one further element (not shown here, see FIG. 2) of the electric cable.

(53) FIG. 14 shows the cable receiving unit 20d in a schematic exploded view.

(54) The cable receiving unit 20d comprises a guiding unit 82d for guiding the conductor cores of the electric cable from the connection unit 28d into the wiring block 22d. The guiding unit 82d connects the connection unit 82d to the wiring block 22d in an angled fashion.

(55) Differently than in the second exemplary embodiment illustrated in FIGS. 8 and 9, the guiding unit 82d is realized in a multi-part implementation. The guiding unit 82d comprises a guiding base body 172d, which is oriented parallel to the plugging direction 14d, and two deflection elements 174d, 176d for an angled connection of the connection unit 28d to the wiring block 22d.

(56) In an assembled state of the plug connector device 10d, the guiding base body 172d is connected to the wiring block 22d in the plugging direction 14d.

(57) The deflection elements 174d, 176d are realized as corresponding half-shells and are configured for establishing a form-fit connection to the guiding base body 172d. The guiding base body 172d has on its rear side, opposed to the plugging direction 14d, a circumferential groove 178d. For the purpose of establishing the form-fit connection, latch edges 180d of the deflection elements 174d, 176d can be slid into the circumferential groove 178d.

(58) The plug connector device 10d comprises a shield contact 40d, which is rotatably connected to the guiding base body 172d and is latchable in at least four positions. In the present case, the shield contact 40d is rotatable by 360°. The guiding base body 172d has four latch grooves (not shown), which are respectively arranged offset from each other by 90° and which the shield contact 40d can be latched into depending on a desired position. Like in the exemplary embodiment depicted in FIGS. 8 and 9, the shield contact 40d is pivotable around a further pivot axis 92d that extends perpendicularly to the plugging direction 14d.

(59) For an assembly of the cable receiving unit **20d**, the shield contact **40d** is connected to the guiding base body **172d**; it is rotated into the desired position and is latched in this position into one of the latch grooves of the guiding base body **172d**. Then the shield contact **40d** is pivoted around the further pivot axis **92d**. After that, the conductor cores of the electric cable are connected to the wiring block **22d**. Then the deflection elements **174d**, **176d**, which are realized as half-shells, are slid with their respective latch edges **180d** into the circumferential groove **178d**.

(60) In the slid-in state, the deflection elements **174d**, **176d** together form a thread **134d**.

Analogously to the preceding exemplary embodiments, the connection unit **28d** includes a further connection element **136d**, which comprises a nut **138d** that corresponds to the thread **134d**. For a fixation of the deflection elements **174d**, **176d** in the slid-in state, the further connection element **136d** is screwed onto the thread **134d**. Analogously to the preceding exemplary embodiment, the connection unit **28d** comprises a sleeve-like connection element **94d**, which is configured to engage around a further element of the cable along a circumferential direction. Differently than in the preceding exemplary embodiment, the sleeve-like connection element **94d** is configured to generate a resistance against a tensile load of the cable perpendicularly to the plugging direction **14d**.

(61) FIG. 13 exemplarily shows the plug connector device **10d** with the cable receiving unit **20d** in an assembled state in a first position, wherein the connection unit **28d** is connected to the wiring block **22d** (see FIG. 14) by means of the guiding unit **82d** in an angled fashion, at an angle of 90°, and is oriented leftwards by means of the guiding unit **82d**. Alternatively, the connection unit **28d** is connectable to the wiring block **22d** by means of the guiding unit **82d** in an angled fashion in at least three further positions, these positions being in each case oriented perpendicularly to the plugging direction **14d**. Relative to the first position shown in FIG. 13, the connection unit **28d** would in a second position be rotated clockwise by 90° and would be oriented upwards, in a third position it would be rotated clockwise by 180° and would be oriented rightwards, and in a fourth position it would be rotated clockwise by 270° and would be oriented downwards (not shown).

REFERENCE NUMERALS

(62) **10** plug connector device **12** plug connector unit **14** plugging direction **16** contact unit **18** electric plug contact **20** cable receiving unit **22** wiring block **24** conductor core **26** electric cable **28** connection unit **30** further element **32** wiring assistance unit **34** plug element **36** housing **38** cable jacket **40** shield contact **42** cable shield **44** first sub-unit **46** second sub-unit **48** flap **50** pivot axis **52** first inner contour **54** first outer contour **56** second inner contour **58** second outer contour **60** first guiding cam **62** first guiding groove **64** second guiding cam **66** second guiding groove **68** width extension **70** width extension **72** circuit board **74** connection line **76** connection line **78** connection line **80** connection line **82** guiding unit **84** contact region **86** connection flap **88** pivot axis **90** portion **92** further pivot axis **94** element **96** lamella **98** direction **100** plug connector kit **102** electric plug contact **104** electric plug contact **106** electric plug contact **108** further electric plug contact **110** further electric plug contact **112** further electric plug contact **114** further electric plug contact **116** further connection line **118** further connection line **120** further connection line **122** further connection line **124** insulation displacement contact **126** first connection path **128** second connection path **130** core sheath **132** connection base body **134** thread **136** further connection element **138** nut **140** slotted link for force transmission **142** outer contour element **144** first method step **146** second method step **148** height extent **150** further circuit board **152** further electric plug contact **154** further electric plug contact **156** further electric plug contact **158** further electric plug contact **160** further electric plug contact **162** further electric plug contact **164** further electric plug contact **166** further electric plug contact **168** EON plug contact **170** metallized pass-through opening **172** guiding base body **174** deflection element **176** deflection element **178** circumferential groove **180** latch edge **182** recess **184** press-fit zone

Claims

1. A plug connector device, in particular an RJ plug connector device, with a plug connector unit which is configured to create a plug connection extending along a plugging direction and which comprises a contact unit with at least two electric plug contacts, the plug connector device comprising a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and comprising a connection unit for creating a connection to at least one further element of the cable that is different from a conductor core, and comprising a wiring assistance unit, which is configured for a tool-less establishing of a connection between the wiring block and the contact unit, wherein the connection unit comprises a strain relief for a cable jacket of the cable, wherein the connection unit comprises a shield contact for contacting a cable shield of the cable to generate a shield contacting before the connection between the wiring block and the contact unit is brought about, wherein the connection unit comprises at least one connection element, wherein the strain relief is provided via the at least one connection element of the connection unit, and wherein the at least one connection element is configured for creating the connection to the cable jacket of the cable.
2. The plug connector device according to claim 1, wherein the plug connector unit comprises a plug element for an insertion into a plug socket.
3. The plug connector device according to claim 1, wherein the wiring assistance unit comprises a first sub-unit, which is part of the plug connector unit, and a second sub-unit which cooperates with the first sub-unit and is part of the cable receiving unit.
4. The plug connector device according to claim 3, wherein the first sub-unit comprises a flap, which is pivotable around a pivot axis extending perpendicularly to the plugging direction and is configured to provide a force acting in the plugging direction for establishing the connection between the wiring block and the contact unit.
5. The plug connector device according to claim 4, wherein the flap has at least one first inner contour, which is configured to cooperate with a first outer contour of the second sub-unit.
6. The plug connector device according to claim 4, wherein the flap has at least one first inner contour, which is configured to cooperate with a first outer contour of the second sub-unit, and which is configured to lock the connection between the wiring block and the contact unit.
7. The plug connector device according to claim 3, wherein the first sub-unit comprises at least one first guiding cam and at least one second guiding cam which is arranged offset from the first guiding cam in a circumferential direction, the guiding cams being configured to engage into corresponding grooves of the second sub-unit.
8. The plug connector device according to claim 7, wherein the guiding cams are arranged off-center with respect to a height extent of the first sub-unit.
9. The plug connector device according to claim 1, including the cable receiving unit comprising the wiring block for an accommodation of conductor cores of the electric cable, and comprising the connection unit for creating a connection to the at least one further element of the cable that is different from a conductor core, wherein the cable receiving unit comprises a guiding unit for a guidance of the conductor cores from the connection unit into the wiring block, the guiding unit connecting the connection unit to the wiring block in an angled fashion.
10. The plug connector device according to claim 9, wherein the guiding unit is realized in multi-part fashion and comprises a guiding base body, which is oriented parallel to the plugging direction, as well as at least two deflection elements for an angled connection of the connection unit to the wiring block.
11. The plug connector device according to claim 10, wherein the deflection elements are realized as corresponding half-shells and are configured for establishing a form-fit connection to the guiding base body.
12. The plug connector device according to claim 10, comprising a shield contact, which is rotatably connected to the guiding base body and is latchable in at least four positions.

13. The plug connector device according to claim 1, including the cable receiving unit, which comprises the wiring block for an accommodation of conductor cores of the electric cable and comprises the connection unit for creating a connection to the at least one further element of the cable which is different from a conductor core, wherein the connection unit comprises at least one sleeve-like connection element, which is configured to engage around the further element of the cable along a circumferential direction and to generate a resistance against a tensile load of the cable counter to the plugging direction.

14. A plug connector kit for producing a plug connector device according to claim 1, with the plug connector unit and with the cable receiving unit.

15. A method for field assembly of an electric cable with a plug connector device according to claim 1, wherein at least one conductor core of the electric cable is connected to at least one electric plug contact of the contact unit.

16. A plug connector device, in particular an RJ plug connector device, with a plug connector unit which is configured to create a plug connection extending along a plugging direction and which comprises a contact unit with at least two electric plug contacts, including a cable receiving unit, which comprises a wiring block for an accommodation of conductor cores of an electric cable and comprises a connection unit for creating a connection to at least one further element of the cable which is different from a conductor core, wherein the connection unit comprises at least one sleeve-like connection element, which is configured to engage around the further element of the cable along a circumferential direction and to generate a resistance against a tensile load of the cable counter to the plugging direction, wherein the connection unit comprises a connection element and a further connection element, which act together for creating the connection to the at least one further element of the cable that is different from a conductor core, and wherein the sleeve like connection element is provided to be arranged between the further element of the cable which is different from a conductor core and the further connection element.

17. A plug connector device, in particular an RJ plug connector device, with a plug connector unit which is configured to create a plug connection extending along a plugging direction and which comprises a contact unit with at least two electric plug contacts, the plug connector device comprising a cable receiving unit comprising a wiring block for an accommodation of conductor cores of an electric cable and comprising a connection unit for creating a connection to at least one further element of the cable that is different from a conductor core, and comprising a wiring assistance unit, which is configured for a tool-less establishing of a connection between the wiring block and the contact unit, wherein the wiring assistance unit comprises a first sub-unit, which is part of the plug connector unit, and a second sub-unit which cooperates with the first sub-unit and is part of the cable receiving unit, wherein the first sub-unit comprises at least one first guiding cam and at least one second guiding cam which is arranged offset from the first guiding cam in a circumferential direction, the guiding cams being configured to engage into corresponding grooves of the second sub-unit.
